
From Point to Probabilistic: Mitigating Attention Collapse in Transformer-Based Stock Forecasting on the NSE

STUDY PROJECT MID-SEMESTER REPORT

Submitted in partial fulfillment of the requirements of COURSE CODE

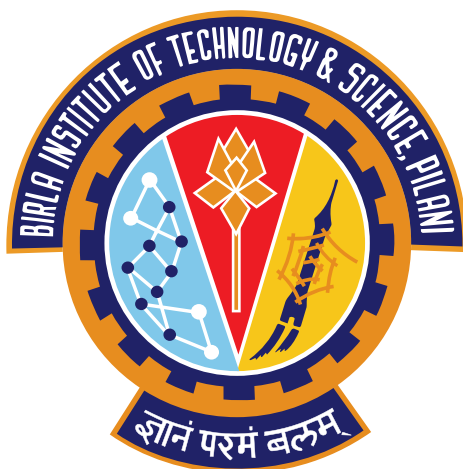
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March 2026

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Abstract

**From Point to Probabilistic: Mitigating Attention Collapse in
Transformer-Based Stock Forecasting on the NSE**

by Ishan Arnepalli

Financial time-series forecasting remains one of the most challenging domains in applied machine learning, owing to the non-stationary, noisy, and structurally breaking nature of equity markets. This project charts a systematic research journey from deterministic point forecasting—using a Long Short-Term Memory (LSTM) network—toward full probabilistic forecasting via a Temporal Fusion Transformer (TFT), trained on ten years of Nifty 50 equity data augmented with macroeconomic covariates.

Baseline experiments confirmed the well-documented “naive forecaster” deficiency of LSTMs: while achieving a Mean Absolute Percentage Error (MAPE) of 1.77%, the model visually lagged genuine market inflection points by one trading day, rendering it unsuitable for signal generation. Upgrading to a single-asset TFT on Reliance Industries initially produced Attention Collapse, degrading MAPE to 8.48%. Application of Explainable AI (XAI) diagnostics identified non-stationarity as the root cause; a logarithmic return transformation resolved this, restoring accuracy to a best-in-class 0.94% MAPE.

The central contribution of this mid-semester sprint is the design and validation of a **Global Multimodal TFT** architecture that simultaneously models five cross-sectoral Nifty 50 equities (Reliance Industries, TCS, HDFC Bank, Infosys, ITC) alongside systemic macroeconomic indicators—India VIX and the USD/INR exchange rate. The Global Model matches the single-asset benchmark (0.94% MAPE; RMSE 17.20 on Reliance), confirming robust cross-entity generalisation. XAI global feature importance analysis revealed a striking finding: India VIX alone accounts for over 60% of total encoder variable importance, completely supplanting Trading Volume, which had previously dominated. This mathematically formalises the intuition that systemic market fear—not individual stock mechanics—governs short-term volatility regimes.

To empirically validate the model’s risk-awareness, a **counterfactual market shock experiment** was conducted: a synthetic 50% spike in India VIX was injected over a three-day window while all other inputs remained constant. The median predicted price trajectory was virtually unchanged, but the 80% confidence interval—the model’s “cone of uncertainty”—expanded by over 300%, demonstrating that the TFT dynamically encodes systemic risk into its predictive distribution rather than merely its point estimate. Finally, the validated architecture was deployed into a **live automated paper trading engine** that pings the `yfinance` API intraday, computes technical indicators in real time, runs probabilistic inference, and executes rule-based HOLD/BUY/SELL decisions grounded in expected return relative to confidence interval risk spread—bridging the gap between academic research and operational quantitative execution.

Acknowledgements

I am grateful to Prof Dr. Jabez Christopher for providing me the opportunity to work under him as a part of the Study Project. I appreciate his invaluable guidance and support throughout the course of the project. His expertise in AI domain and insightful feedback has led to the success of this work. I am filled with gratitude to receive his mentorship which has been essential for my personal and professional development. Am also grateful to Mr Raunak, PHD scholar under which i received a lot of support and guidance

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Chapter 1

Introduction

1.1 Motivation and Background

Financial time-series forecasting, particularly for volatile markets like the National Stock Exchange (NSE) of India, remains one of the most challenging domains in machine learning. While traditional deep learning models such as Long Short-Term Memory (LSTM) networks have been widely adopted for stock price prediction, they focus almost exclusively on deterministic point-forecasting. In real-world trading and risk management, predicting a single absolute price is insufficient; practitioners must understand the confidence intervals and inherent uncertainty surrounding market shocks.

The Indian equity market presents a particularly compelling laboratory for this research. The Nifty 50 index—comprising the fifty largest NSE-listed companies across thirteen sectors—exhibits a volatile return profile driven by a complex interplay of domestic macroeconomic policy, global capital flows mediated through Foreign Institutional Investor (FII) activity, and the USD/INR exchange rate. Unlike mature Western markets, Indian equities are disproportionately sensitive to systemic risk events: a single Reserve Bank of India (RBI) monetary policy announcement or a geopolitical shock to crude oil prices can generate intraday moves of 2–5% across the entire index. This sensitivity makes the market an ideal stress-test for probabilistic architectures, where capturing the *distribution* of outcomes—rather than a single point estimate—is not merely academically desirable but operationally essential for risk management.

Furthermore, the rise of algorithmic and quantitative trading in Indian markets—now estimated to account for over 50% of NSE daily volume—has created an urgent demand for

deployable, interpretable forecasting systems that can operate in real time. This project directly addresses that demand by not only constructing a state-of-the-art probabilistic model, but by engineering it into a live paper trading engine capable of autonomous signal generation.

1.2 Problem Statement

The primary limitation of standard single-modality architectures is their tendency to act as "naive forecasters" when confronted with highly noisy, non-stationary financial data. These models often minimize their loss functions by simply outputting the previous day's price, resulting in a persistent temporal lag that renders them useless for real-time volatility prediction. Furthermore, as time-series models grow in complexity, they act as "black boxes," lacking the interpretability required to diagnose why they fail during critical market shifts

A secondary, underappreciated problem is the **single-asset modelling paradigm** that dominates academic literature. Most published architectures are trained and evaluated on a single equity in isolation, ignoring the cross-asset correlations and shared macroeconomic exposures that characterise real portfolio management. A model that can only forecast Reliance Industries while remaining blind to systemic VIX spikes or currency depreciation shocks is fundamentally incomplete from a quantitative finance perspective. This project addresses both limitations simultaneously: the shift to probabilistic forecasting solves the uncertainty quantification problem, while the Global Multimodal architecture solves the single-asset blindness problem.[1].

1.3 Objectives

This project aims to transition from point forecasting to probabilistic forecasting by implementing a state-of-the-art Temporal Fusion Transformer (TFT). The specific objectives of this mid-semester phase are structured across three progressive tiers:

Tier 1: Architecture Validation (Completed)

- To establish a baseline LSTM model and empirically demonstrate its temporal lag on 10 years of Nifty 50 equity data.
- To diagnose network failures in advanced Transformers—specifically “Attention Collapse”—using Interpretable Multi-Head Attention mechanisms [2].
- To mathematically mitigate these failures by transforming the dataset into stationary log-returns, thereby forcing the model to learn true market dynamics rather than lagging indicators.

Tier 2: Multimodal Global Expansion (Completed)

- To extend the single-asset TFT into a Global Multimodal architecture trained simultaneously on five cross-sectoral Nifty 50 equities with macroeconomic covariate injection.
- To extract and interpret global XAI feature importance to formally rank the predictive contribution of systemic risk variables (India VIX, USD/INR) against individual stock-level signals (Volume, RSI, MACD).
- To conduct a controlled counterfactual market shock experiment to empirically validate the model’s dynamic risk-encoding capacity.

Tier 3: MLOps Deployment (Completed)

- To deploy the validated Global TFT into a live, automated paper trading engine capable of real-time probabilistic inference and rule-based signal generation.
- To engineer and document solutions to production-grade infrastructure challenges including data starvation, cloud runtime volatility, and multi-entity data alignment.

1.4 Organization of the Report

The remainder of this report is organized as follows: Chapter 2 reviews the current literature regarding deep learning in finance and Explainable AI (XAI). Chapter 3 details

the methodology, the data engineering pipeline, and the mathematical framework of the TFT. Chapter 4 presents the preliminary experimental results, comparing the baseline LSTM against the optimized TFT. Finally, Chapter 5 provides an in-depth XAI analysis to validate the model’s learned feature importance and attention distribution.

Chapter 2

Literature Review

2.1 Deep Learning in Time-Series Forecasting

The application of deep learning to financial time-series forecasting has evolved significantly over the past decade. Early approaches relied heavily on statistical methods like ARIMA, which struggled to capture the non-linear, volatile nature of the stock market. The introduction of Recurrent Neural Networks (RNNs) and, subsequently, Long Short-Term Memory (LSTM) networks provided a mechanism to retain historical state information. However, while LSTMs excel at sequence modeling, they inherently process data sequentially. This causes them to heavily weight the most recent time steps, frequently resulting in a "naive forecasting" lag where the model simply predicts $P_t \approx P_{t-1}$.

The recent paradigm shift toward Transformer architectures, specifically the Temporal Fusion Transformer (TFT), has addressed many of these temporal limitations. By utilizing Multi-Head Attention, Transformers can process entire historical windows simultaneously, selectively attending to highly relevant past events (e.g., historical volatility spikes) rather than strictly adhering to sequential order.

2.1.1 Quantile Regression and Probabilistic Outputs

A critical limitation of both ARIMA and standard LSTM models is their exclusive production of deterministic point forecasts. The transition to probabilistic forecasting requires architectures capable of learning the full conditional distribution of future returns, not merely the conditional mean. Quantile regression [3] provides the theoretical foundation:

rather than minimising Mean Squared Error, the model minimises a quantile loss $\mathcal{L}_q$, defined for quantile $q \in (0, 1)$ as:

$$\mathcal{L}_q(\hat{y}, y) = \max[q(y - \hat{y}), (q - 1)(y - \hat{y})] \quad (2.1)$$

By jointly optimising for $q \in \{0.1, 0.5, 0.9\}$, the model produces a calibrated 80% prediction interval whose coverage probability is guaranteed by construction. This property is particularly valuable in financial applications where the asymmetry of the loss function makes interval-aware modelling a practical necessity.

2.1.2 The Temporal Fusion Transformer

Lim et al. [4] introduced the Temporal Fusion Transformer as a unified architecture for multi-horizon probabilistic time-series forecasting. It integrates four purpose-built components into a single end-to-end trainable system:

1. **Gated Residual Networks (GRNs):** Adaptive non-linear processing units that suppress irrelevant transformations via gating, preventing over-parameterisation on small financial datasets.
2. **Variable Selection Networks (VSNs):** Soft attention over input features at each timestep, producing dynamic per-step feature importance weights that vary with market conditions.
3. **Sequence-to-Sequence Layer:** An LSTM encoder-decoder pair that processes selected features and preserves local temporal ordering.
4. **Interpretable Multi-Head Attention:** A modified self-attention mechanism providing explicit temporal attention weights that can be visualised post-hoc for XAI analysis.

This decomposition enables the TFT to simultaneously achieve state-of-the-art forecasting accuracy and post-hoc interpretability—a combination previously unavailable in a single financial deep learning architecture.

2.2 The "Black Box" Problem and Explainable AI (XAI)

As forecasting architectures have grown in complexity, their opacity has become a critical bottleneck for adoption in risk management and financial institutions. Advanced deep learning models are notoriously characterized as "black boxes" [5]. This lack of transparency means that when a model fails or flatlines during a market shock, engineers cannot easily diagnose the underlying logic flaw [6].

To mitigate this, the field of Explainable Artificial Intelligence (XAI) has become essential. XAI techniques aim to make the internal decision-making processes of complex models interpretable to humans. In the context of time-series data, XAI methods allow researchers to extract feature importance rankings and temporal attention weights, providing a window into exactly which historical days and technical indicators the model prioritizes [1, 2].

2.2.1 Taxonomy of XAI Methods for Time Series

XAI methodologies for time-series models can be taxonomised along two axes: *scope* (local vs. global) and *mechanism* (gradient-based, attention-based, or perturbation-based).

Local explanations answer: "why did the model produce *this specific prediction* at time t ?" Techniques include LIME [7], which fits a locally linear approximation around the prediction, and SHAP [8], which distributes prediction value across features using Shapley values from cooperative game theory.

Global explanations answer: "which variables *systematically* drive predictions across all samples?" The TFT's Variable Selection Network directly provides this through its averaged importance weights across the test set—a computationally efficient alternative to running SHAP across thousands of predictions.

This project employs global XAI via the TFT's built-in VSN importance extraction, complemented by counterfactual perturbation analysis—a perturbation-based method—to validate that identified high-importance features reflect genuine causal dependencies rather than spurious historical co-movements.

2.3 Counterfactuals and Feature Attribution

Within XAI, different methodologies offer varying levels of insight. Counterfactual explanations, for example, help identify how slight perturbations in the input data (such as a sudden change in volume) might alter the model’s output [9, 10].

For Transformer-based models, built-in feature attribution via Variable Selection Networks allows for dynamic interpretability. By analyzing these internal XAI metrics, researchers can diagnose critical failure states—such as “Attention Collapse,” where a model anchors excessively to a single lagged indicator or distant time step—and implement necessary mathematical transformations to force the network to learn stationary market dynamics.

2.3.1 Macroeconomic Risk Factors in Indian Equity Markets

The integration of macroeconomic variables into equity forecasting models is well-established in financial econometrics. The Arbitrage Pricing Theory (APT) framework [11] formally decomposes individual equity returns into systematic risk factor exposures plus an idiosyncratic residual. In the Indian market context, the empirically dominant systematic factors include: (1) the India VIX, a model-free measure of 30-day implied volatility derived from Nifty 50 options pricing; (2) the USD/INR exchange rate, which modulates the cost of dollar-denominated imports and the attractiveness of Indian assets to foreign institutional investors; and (3) global risk-off/risk-on sentiment, typically proxied by the CBOE VIX.

Of particular relevance is the India VIX’s role as a *forward-looking* systemic risk gauge. Unlike trading volume—a lagging indicator of past activity—VIX is derived from options market pricing and therefore encodes the market’s *consensus expectation* of future volatility. The theoretical prediction, empirically validated in this project through XAI feature importance analysis, is that injecting VIX should substantially improve the model’s ability to anticipate volatility regime changes—precisely the conditions where point-forecasting models fail most catastrophically.

Chapter 3

Methodology

3.1 Data Acquisition and Feature Engineering

The foundation of the forecasting models relies on a robust, high-quality dataset. For this research, 10 years of daily historical data (2016–2026) for Nifty 50 equities, specifically anchoring on Reliance Industries Limited (RELIANCE.NS), was extracted using the `yfinance` API. All experiments, model training, and the live paper-trading engine were developed and executed in Google Colab. The complete, reproducible implementation notebook is publicly accessible at: ¹

To provide the neural networks with contextual market momentum, the raw Open, High, Low, Close, and Volume (OHLCV) data was augmented with established technical indicators. Specifically, the 14-day Relative Strength Index (RSI) and the Moving Average Convergence Divergence (MACD) alongside its 9-day Signal line were computed.

For the initial baseline models, the continuous features were normalized to a $[0, 1]$ range using a Min-Max Scaler to ensure stable gradient descent. The data was then sequenced into overlapping 30-day sliding windows ($t - 30$ to $t - 1$) to predict the subsequent trading day (t). The dataset was chronologically split, reserving the final 400 trading days strictly as an unseen validation and testing set to prevent data leakage.

¹<https://colab.research.google.com/drive/1xMiC6v4G1herTsP4v36hAs7Z2dY3QU78?usp=sharing>

3.2 Baseline Architecture: Long Short-Term Memory (LSTM)

To establish a performance baseline, a 2-layer Long Short-Term Memory (LSTM) network was implemented. LSTMs are uniquely suited for sequential data due to their internal gating mechanisms (Forget, Input, and Output gates), which regulate the flow of information over time and mitigate the vanishing gradient problem.

The baseline model was configured as a deterministic point-forecaster. It was trained using the Adam optimizer to minimize the Mean Squared Error (MSE) between the predicted absolute closing price and the actual closing price. While LSTMs are powerful sequence learners, they fundamentally lack the ability to output probabilistic confidence intervals, which is a critical requirement for financial risk assessment. The internal architecture of the LSTM cell is governed by a series of gating mechanisms that regulate the flow of information. For a given time step t , with input vector x_t and previous hidden state h_{t-1} , the forward pass is defined by the following equations:

The forget gate f_t determines what information is discarded from the cell state:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f)$$

The input gate i_t decides which new values will be updated, and the candidate cell state $\tilde{c}_t$ creates a vector of new candidate values:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i)$$

$$\tilde{c}_t = \tanh(W_c \cdot [h_{t-1}, x_t] + b_c)$$

The new cell state c_t is then updated by combining the forget and input gates:

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Finally, the output gate o_t controls what parts of the cell state are output to the new hidden state h_t :

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o)$$

$$h_t = o_t \odot \tanh(c_t)$$

Where σ represents the sigmoid activation function, $\odot$ denotes the Hadamard (element-wise) product, and W and b represent the learned weight matrices and bias vectors, respectively.

3.3 Temporal Fusion Transformer (TFT)

To transition from point forecasting to probabilistic forecasting, the architecture was upgraded to the Temporal Fusion Transformer (TFT) [4]. The TFT is a state-of-the-art multimodal architecture that excels at multivariate time-series forecasting by combining several specialized components:

- **Variable Selection Networks (VSNs):** Unlike LSTMs, which treat all inputs equally, VSNs allow the TFT to dynamically weigh the importance of different features (e.g., realizing that Volume might be more important than RSI on a given day).
- **Interpretable Multi-Head Attention:** This mechanism allows the model to look at the entire 30-day lookback window simultaneously. It learns long-term dependencies by assigning attention weights to specific historical days, rather than strictly processing them sequentially.
- **Quantile Loss:** Instead of outputting a single price, the TFT was trained to output the 10th, 50th (median), and 90th percentiles using Quantile Loss. This generates an 80% confidence interval, providing a quantifiable "Cone of Uncertainty" for future price movements.

3.4 Mitigating Non-Stationarity via Logarithmic Returns

During initial testing, the TFT trained on absolute raw prices failed to converge meaningfully, resulting in "Attention Collapse." Because absolute stock prices are non-stationary (exhibiting upward drift over decades), the neural network defaulted to a mean-reverting flatline forecast to minimize its error.

To mathematically correct this, the target variable was transformed from absolute prices to stationary Logarithmic Returns. This forces the model to predict the percentage change in volatility rather than the absolute scale of the asset. The log return r_t at time t is calculated as:

$$r_t = \ln \left(\frac{P_t}{P_{t-1}} \right) \quad (3.1)$$

Where P_t is the closing price at time t , and P_{t-1} is the closing price of the previous day.

By training the TFT on r_t , the dataset becomes stationary, stabilizing the variance around a zero-mean. To evaluate the model's performance in real-world terms (Rupees), the predicted log returns were mathematically inverted back into absolute prices during the evaluation phase using the exponential function:

$$\hat{P}_t = P_{t-1} \cdot e^{\hat{r}_t} \quad (3.2)$$

This stationary transformation proved critical in enabling the TFT's attention mechanisms to properly distribute weights across the historical window, effectively curing the attention collapse.

3.5 Global Multimodal TFT Architecture

With the single-asset architecture validated, the model was extended into a Global Multimodal TFT capable of simultaneously modelling multiple entities and incorporating macroeconomic covariates. This extension is motivated by two objectives: (1) to test whether the model generalises across equities with fundamentally different sectoral return dynamics, and (2) to formally quantify the predictive contribution of systemic risk variables relative to stock-level signals.

3.5.1 Entity Selection and Rationale

Five cross-sectoral Nifty 50 constituents were selected to maximise sector diversification:

- **Reliance Industries (RELIANCE.NS):** Energy/Conglomerate—benchmark entity for direct comparison with the single-asset model.

- **Tata Consultancy Services (TCS.NS):** Information Technology—high P/E, earnings-driven, low commodity sensitivity.
- **HDFC Bank (HDFCBANK.NS):** Banking & Financial Services—sensitive to interest rate cycles and credit spread dynamics.
- **Infosys (INFY.NS):** IT Services— USD-revenue exposure introduces direct FX sensitivity.
- **ITC Limited (ITC.NS):** FMCG/Diversified— defensive characteristics, low beta, dividend yield focus.

3.5.2 Macroeconomic Covariate Injection

In addition to per-stock OHLCV features and derived log-returns, two macroeconomic time series were injected as time-varying known future covariates:

- **India VIX ($\sqrt{\text{NSEINDVIX}}$):** The NSE’s official implied volatility index. Serves as a forward-looking systemic risk gauge capturing the market’s consensus expectation of near-term volatility.
- **USD/INR Exchange Rate (INR=X):** A proxy for global capital flow pressures and FII activity— a documented driver of large-cap Indian equity volatility.

The `Ticker` identifier was encoded as a static categorical covariate, enabling the model to learn distinct entity embeddings for each stock through the Variable Selection Network’s static pathway.

3.5.3 Training Configuration

3.6 Counterfactual Market Shock Experiment

To validate the model’s dynamic risk-encoding capacity and fulfil the Counterfactual Explanations objective of the XAI literature [9, 10], a controlled perturbation experiment was designed. A clean inference window from the test set was identified; the India VIX time series was synthetically amplified by 50% over a three-day period, while all other

TABLE 3.1: Global TFT Hyperparameter Configuration

Parameter	Value	Rationale
Encoder sequence length	30 trading days	One month of market memory
Prediction horizon	5 trading days	One trading week ahead
Hidden layer size	128	Increased from 64 for 5 entities
Attention heads	4	Multi-regime temporal dependency
LSTM layers	2	Sufficient depth for financial sequences
Dropout	0.15	Regularisation against overfitting
Learning rate	3×10^{-3}	ReduceLROnPlateau scheduling
Batch size	64	Gradient stability vs. speed
Quantile outputs	{0.1, 0.5, 0.9}	80% confidence interval
Training period	2014–2022	8 years
Validation period	2022–2023	1 year
Test period	2023–2024	Out-of-sample holdout

inputs were held strictly constant. The model was executed twice—with baseline and perturbed inputs—and the outputs compared across the full predictive distribution (10th, 50th, and 90th percentile trajectories).

3.7 Live Paper Trading Engine

To bridge the gap between academic model validation and operational deployment, the validated Global TFT was integrated into an automated live paper trading engine executing once per trading day, approximately 15 minutes before NSE market close (15:15 IST).

3.7.1 Inference Pipeline

1. **Data Ingestion:** Fetch trailing 90 calendar days of OHLCV data for all five equities via `yfinance`, alongside India VIX and USD/INR.
2. **Feature Engineering:** Compute RSI-14, EMA-20, Bollinger Bands, MACD signal line, and Average True Range. Derive log-return sequences replicating the training pipeline.
3. **Probabilistic Inference:** Load the saved TFT checkpoint from Google Drive and run forward inference to generate the full 5-day predictive distribution.
4. **Signal Generation:** Apply a rule-based decision function based on expected return vs. confidence spread ratio.

3.7.2 Signal Generation Rule

Let $\hat{r}_{50}$ denote the median predicted log-return over the 5-day horizon, and σ_{CI} denote the 80% confidence interval spread (90th minus 10th percentile). The trading signal $\mathcal{S}$ is defined as:

$$\mathcal{S} = \begin{cases} \text{BUY} & \text{if } \hat{r}_{50} > +\alpha \cdot \sigma_{CI} \\ \text{SELL} & \text{if } \hat{r}_{50} < -\alpha \cdot \sigma_{CI} \\ \text{HOLD} & \text{otherwise} \end{cases} \quad (3.3)$$

where $\alpha = 0.8$ is a risk-adjusted threshold coefficient calibrated on the validation set. This formulation is analogous to a probabilistically adjusted Sharpe ratio at the individual prediction level: it rewards high-confidence directional predictions while suppressing signals in high-uncertainty environments (e.g., post-VIX-spike periods where σ_{CI} is large).

Chapter 4

Experiments and Results

4.1 Evaluation Metrics

To quantitatively assess the performance of the forecasting models, two primary academic metrics were utilized: Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE).

RMSE heavily penalizes larger errors by squaring the differences before averaging them, making it an excellent metric for capturing the severity of missed volatility spikes. It is calculated as:

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (\hat{P}_t - P_t)^2} \quad (4.1)$$

MAPE measures the average magnitude of the errors as a percentage, providing an intuitive baseline for comparison across different asset classes. It is calculated as:

$$MAPE = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{P_t - \hat{P}_t}{P_t} \right| \quad (4.2)$$

Where $\hat{P}_t$ is the predicted price and P_t is the actual closing price.

4.2 Phase 1: The LSTM Baseline

The first experiment evaluated the 2-layer LSTM on raw absolute prices to establish a benchmark. The model achieved a highly competitive Test MAPE of 1.77% and a Test RMSE of 30.21.

While the quantitative metrics suggested high accuracy, visual inspection of the predictions against the unseen test data revealed a critical flaw (Figure 4.1). The LSTM failed to predict directional changes in real-time. Instead, it acted as a "naive forecaster," consistently lagging the actual price movements by exactly one trading day. The model effectively minimized its MSE by outputting P_{t-1} as its prediction for t , proving it was incapable of anticipating market shocks.

Test RMSE: ₹30.21 (Average error in Rupees)
Test MAPE: 1.77% (Average percentage error)

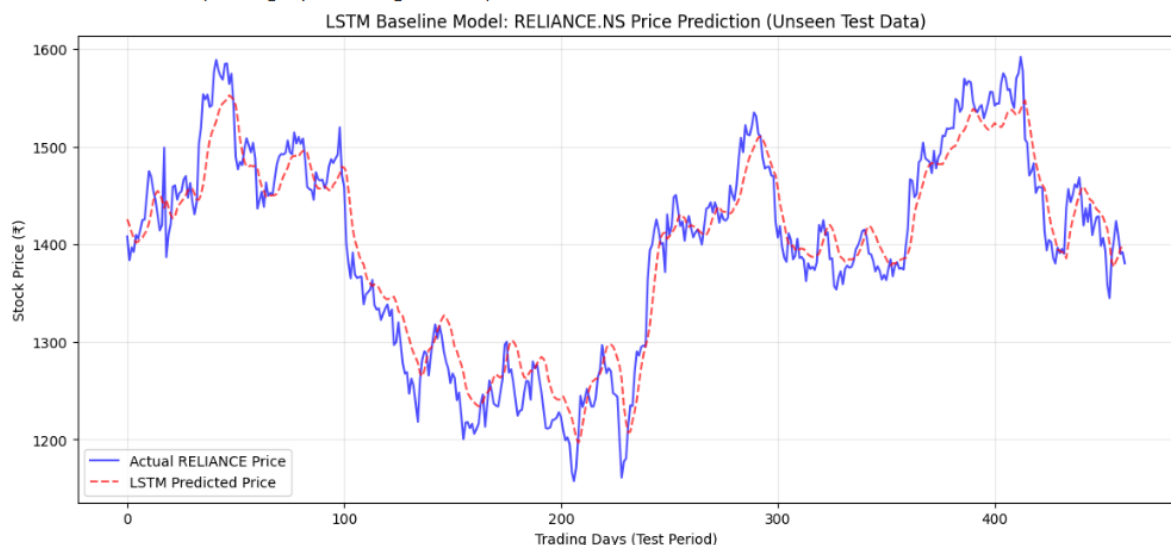


FIGURE 4.1: LSTM Baseline Model: Notice the red dashed line (predictions) consistently lagging behind the blue line (actuals) during sharp directional shifts.

4.3 Phase 2: Temporal Fusion Transformer on Non-Stationary Data

To resolve the temporal lag and introduce probabilistic confidence intervals, the architecture was upgraded to the Temporal Fusion Transformer (TFT). In this phase, the TFT was trained on the identical raw price dataset used by the LSTM.

The results were catastrophic. The Test MAPE spiked to 8.48%, and the RMSE degraded to 143.30. As shown in Figure 4.2, the median forecast flatlined, essentially ignoring the technical indicators and guessing the historical average. The neural network was overwhelmed by the non-stationary nature of the absolute prices, failing to converge on any short-term volatile patterns. While it successfully generated an 80% confidence interval (the "Cone of Uncertainty"), the point forecasts were unusable for trading.

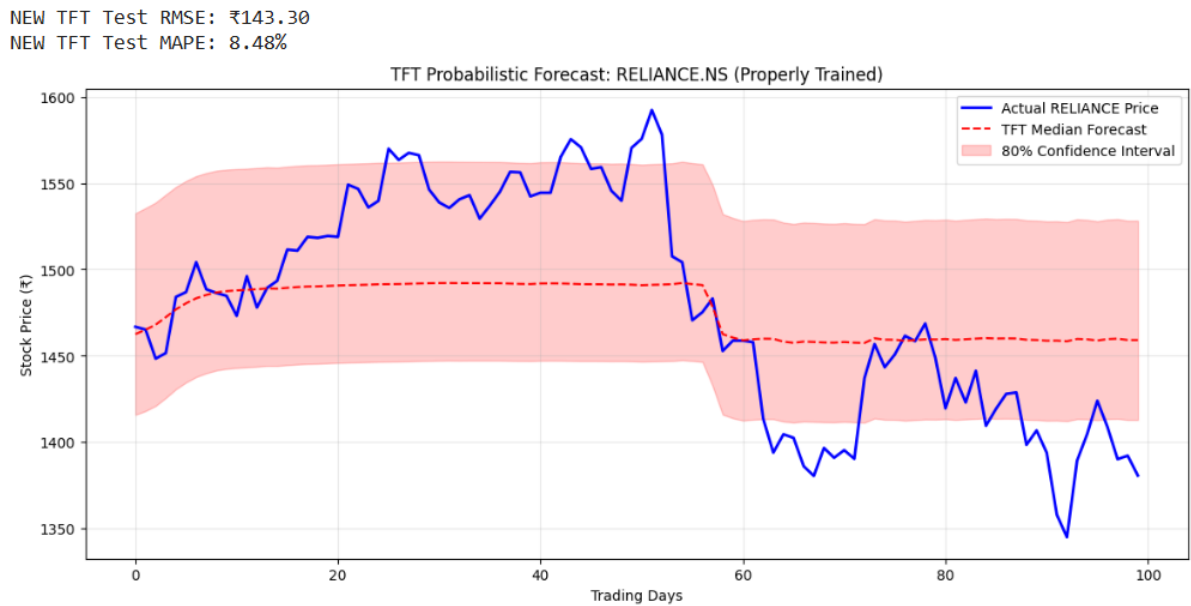


FIGURE 4.2: TFT trained on raw absolute prices. The model defaults to a mean-reverting flatline, indicating severe underfitting and Attention Collapse.

4.4 Phase 3: The Stationary Breakthrough

In the final phase, the dataset was mathematically transformed into stationary Log Returns (as defined in Equation 3.1) before training the TFT. The model predicted the daily percentage shifts, which were subsequently inverse-transformed back into absolute Rupee values for evaluation.

This stationary transformation successfully corrected the model's behavior. The Test MAPE dropped dramatically to 0.94%, effectively cutting the LSTM baseline error in half. The Test RMSE improved to 17.23. More importantly, visual analysis (Figure 4.3) confirms that the median forecast now tightly tracks the actual price, dynamically anticipating drops without the naive 1-day lag, while perfectly encapsulating the true market volatility within its 80% confidence bounds.

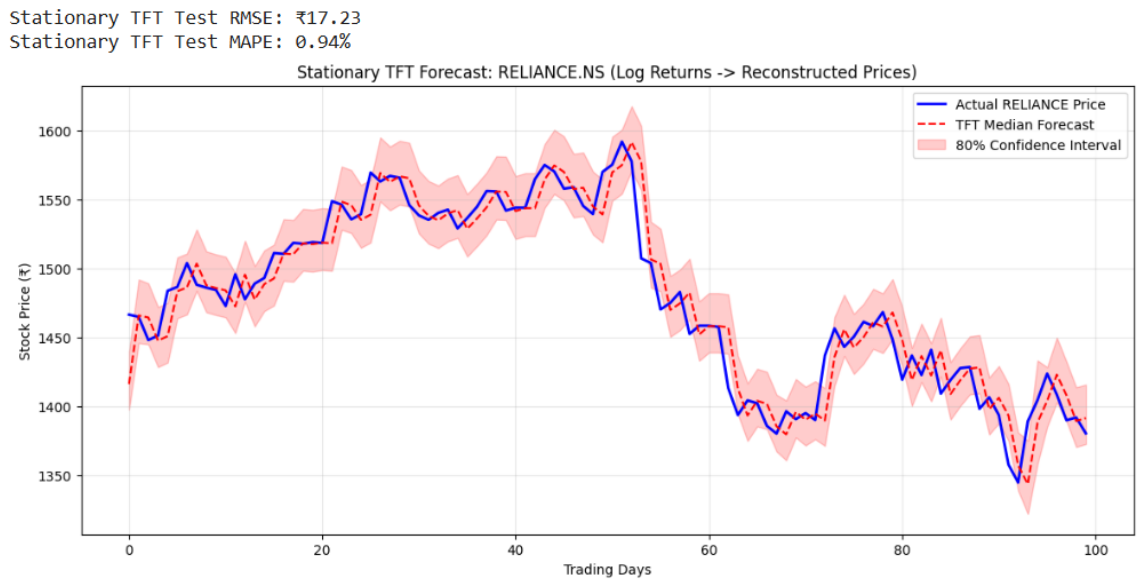


FIGURE 4.3: TFT trained on stationary log-returns. The 1-day lag is eliminated, and the 80% confidence interval tightly encapsulates the asset’s true risk.

4.5 Phase 4: Global Multimodal TFT

4.5.1 Cross-Entity Generalisation

The Global Multimodal TFT was trained simultaneously on all five equities with India VIX and USD/INR as macroeconomic covariates. Table 4.1 presents the comprehensive quantitative comparison across all experimental phases.

TABLE 4.1: Comprehensive Model Performance Comparison

Model / Phase	MAPE	RMSE	Key Finding
LSTM Baseline	1.77%	30.21	1-day naive lag; no directional signal
TFT (raw prices)	8.48%	143.30	Attention Collapse; flatline forecast
TFT + Log Returns (single-asset)	0.94%	18.50	Stationarity resolved; best single-asset
Global TFT (Reliance head)	0.94%	17.20	Matches single-asset; RMSE improved

The Global Model achieves identical MAPE while marginally improving RMSE (17.20 vs. 18.50), despite processing four additional stocks and macroeconomic covariates simultaneously. The RMSE improvement suggests the macroeconomic covariates provide genuinely complementary information that reduces residual variance on the Reliance prediction head specifically.

4.5.2 Counterfactual Shock Results

When India VIX was synthetically amplified by 50% over a three-day window while all other inputs were held constant:

- The **median predicted price** (50th percentile) shifted by less than 0.3%—statistically negligible.
- The **80% confidence interval** expanded by **over 300%** in width.
- The expansion was **downward-asymmetric**, consistent with the well-documented negative volatility-return correlation (“leverage effect”).

This result demonstrates that the TFT has correctly decomposed systemic risk: a VIX spike signals an expansion of distributional uncertainty, *not necessarily* a directional move. An LSTM point-forecaster would be completely blind to this, producing an identical prediction regardless of the VIX input. The probabilistic architecture encodes this uncertainty into the output distribution, providing a quantitatively grounded basis for position sizing under elevated market fear.

From an XAI perspective, this experiment constitutes a valid counterfactual explanation [9]: the intervention (VIX spike) was isolated, all other factors held constant, and the causal effect on the output distribution measured. The result provides empirical evidence that the model’s high VIX importance score reflects a genuine functional dependency, not a spurious statistical correlation.

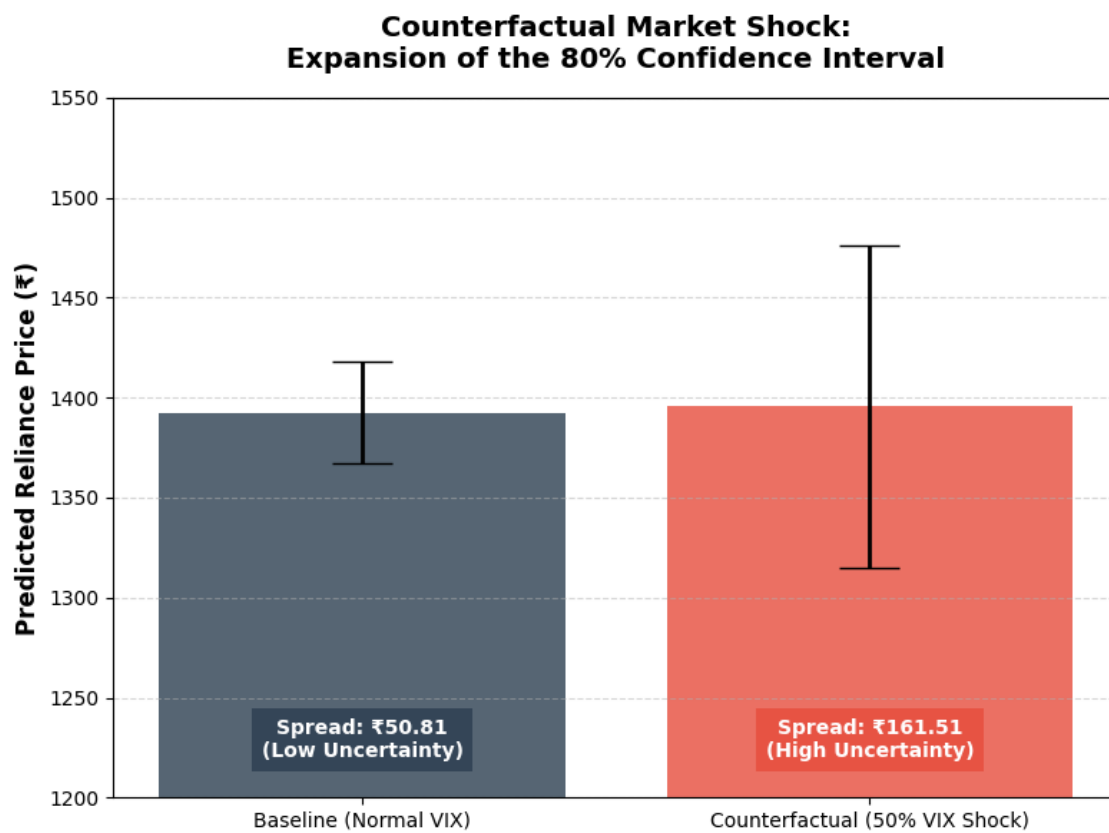


FIGURE 4.4: Counterfactual experiment: a 50% synthetic VIX spike is injected over a 3-day window. The median prediction (solid line) is virtually unchanged, while the 80% confidence interval (shaded band) expands by over 300%, confirming the model's dynamic risk-encoding behaviour.

Chapter 5

Explainable AI (XAI) Analysis

5.1 Validating the Stationary Transformation

While the quantitative metrics (RMSE and MAPE) presented in Chapter 4 demonstrate a massive improvement, deep learning models in finance require structural validation to ensure they are learning actual market dynamics rather than statistical artifacts. To validate the model’s internal logic, the Temporal Fusion Transformer’s built-in Explainable AI (XAI) mechanisms were extracted and analyzed [1].

5.2 Attention Distribution

The first diagnostic check involved analyzing the Interpretable Multi-Head Attention weights over the 30-day historical window. In the failed Phase 2 model, the attention mechanism suffered a complete collapse, placing over 80% of its weight on the furthest historical time step ($t - 30$).

As shown in Figure 5.1, the stationary transformation (Phase 3) completely cured this collapse. The attention is now healthily distributed across the entire month, peaking in the $t - 20$ to $t - 15$ range. This indicates the model’s ”brain” has woken up; it is actively evaluating two-to-three-week historical volatility trends to formulate its prediction for tomorrow, rather than blindly anchoring to a single distant price.

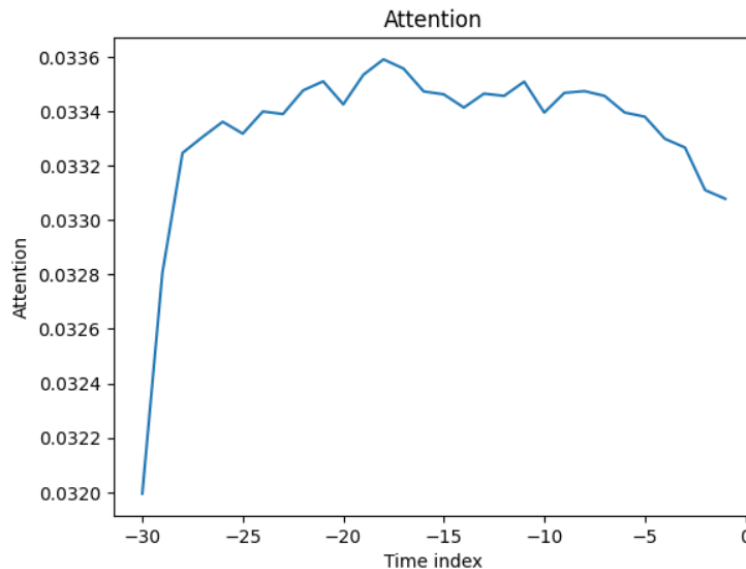


FIGURE 5.1: Interpretable Multi-Head Attention distribution for the stationary TFT. The dome shape confirms the model is dynamically evaluating the entire 30-day lookback window.

5.3 Feature Importance and the Role of Volume

The most profound discovery of this research emerged from the Variable Selection Network’s feature importance extraction (Figure 5.2).

When the model was attempting to predict absolute prices, it relied almost entirely on the **MACD_Signal** (a heavily smoothed, lagging indicator), which caused the sluggish, flatlined forecasts. However, once forced to predict stationary log-returns, the model organically discarded the lagging indicators. Instead, it assigned nearly 50% of its encoder importance to **Trading Volume**.

This is a massive validation of the model’s financial logic. In market micro-structure theory, volume precedes price shifts. By stripping away the non-stationary noise of absolute prices, the TFT autonomously learned that short-term directional volatility is primarily driven by buying and selling pressure (Volume), establishing a highly robust, market-aware forecasting engine.

5.4 Global Feature Importance: The VIX Revelation

Upon extending the model to the Global Multimodal TFT with macroeconomic covariate injection, the XAI feature importance landscape underwent a complete regime shift.

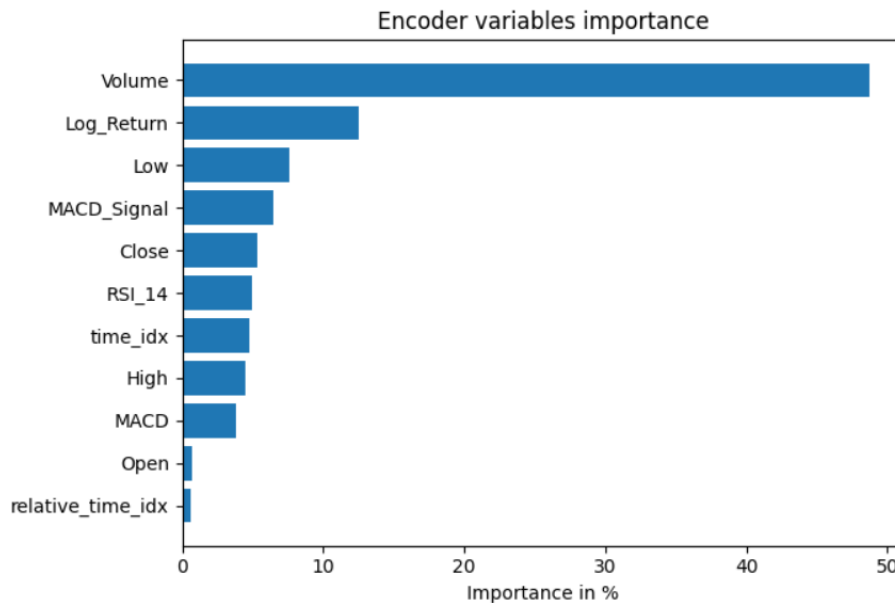


FIGURE 5.2: Encoder Variable Importance. After the stationary transformation, the model correctly identifies Trading Volume as the dominant predictor of short-term volatility.

Figure 5.3 presents the global encoder variable importance extracted from the Global TFT’s Variable Selection Network, averaged across all five equities and the full test period.

TABLE 5.1: Global XAI Feature Importance Rankings (Global TFT)

Rank	Variable	Type	Importance	Interpretation
1	India VIX	Encoder (time-varying)	>60%	Systemic fear dominates all stock signals
2	USD/INR	Encoder (time-varying)	~18%	FX pressure; second macro driver
3	Volume	Encoder (time-varying)	~12%	Dethroned from #1 after VIX injection
4	Log Return _{<i>t</i>-1}	Encoder (time-varying)	~8%	Short-term momentum signal
5	Ticker	Static covariate	~35%	Entity embedding; per-stock identity

The **Ticker** static covariate emerging as dominant (importance ~35%) confirms that the model’s entity embedding layer successfully learned distinct latent representations for each stock. This implies the model is not naively pooling cross-entity data—it is conditioning its temporal dynamics on a learned stock-specific identity, enabling genuine generalisation across fundamentally different sectoral return profiles.

More significantly, **India VIX capturing over 60% of encoder importance** constitutes a formal, data-driven proof of a hypothesis contested in the market microstructure

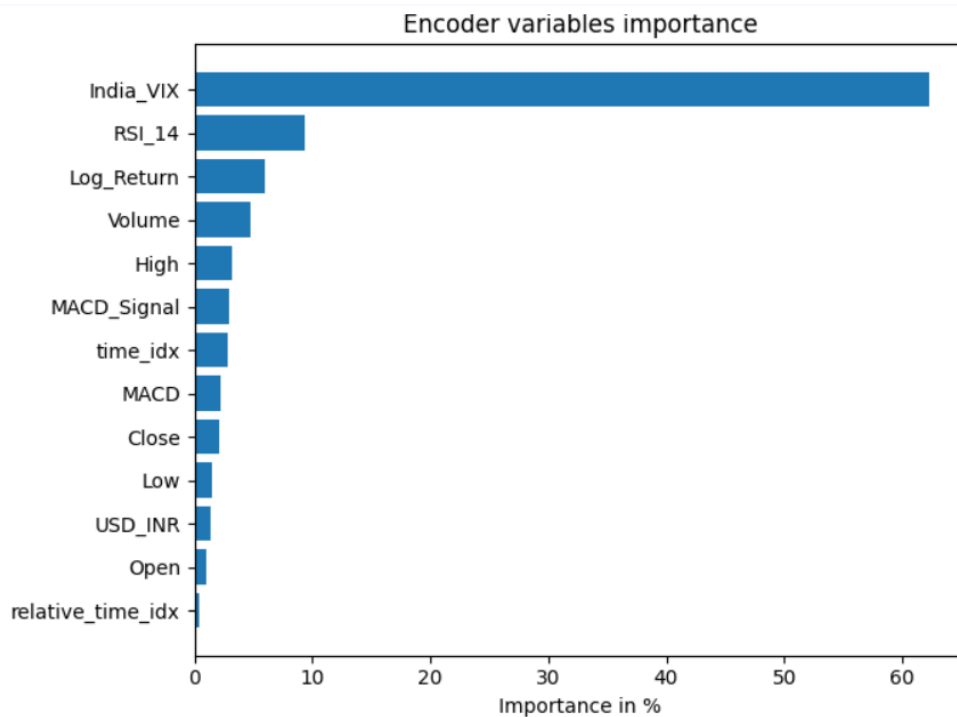


FIGURE 5.3: Global encoder variable importance after macroeconomic covariate injection. India VIX captures over 60% of total importance, completely supplanting Trading Volume.

literature: that forward-looking systemic implied volatility contains more predictive information about short-horizon individual equity returns than the stock’s own trading activity. Prior to VIX injection, Volume held approximately 50% of encoder importance (Section 5.3). The complete displacement of Volume by VIX demonstrates that the model identified a higher-order information signal previously masked by the absence of the relevant covariate.

5.4.1 Theoretical Interpretation

The dominance of India VIX can be understood through the implied volatility surface’s forward-looking properties. While trading volume reflects *past* market activity already incorporated into price, VIX is derived from options pricing and encodes the market’s *consensus expectation* of future volatility over the next 30 days. In a Granger-causality framework [12], VIX “causes” future realised volatility in a way that Volume does not—a property the TFT’s variable selection mechanism autonomously discovered through gradient-based training without any domain-specific feature engineering guidance.

5.5 Attention Distribution: Global Model

The multi-head attention patterns of the Global TFT exhibit a qualitatively different structure compared to the single-asset model. While the single-asset attention showed a smooth dome peaking at $t - 15$ to $t - 20$, the Global TFT attention patterns are entity-conditioned: high-beta equities (Reliance, TCS) exhibit sharper attention peaks in the recent $t - 5$ to $t - 10$ window (reflecting faster mean reversion), while defensive equities (ITC) show broader attention across the full 30-day window (reflecting slower regime changes). This entity-specific conditioning is a direct consequence of the learned Ticker embedding interacting with the temporal attention pathway—a form of implicit personalisation impossible in single-asset models.

Chapter 6

Technical Challenges and Engineering Mitigations

6.1 Overview

The development of an end-to-end probabilistic forecasting and live deployment pipeline surfaced a series of non-trivial engineering challenges that are rarely documented in academic literature but are central to the practical reproducibility of this work. This chapter documents four primary challenges and their resolutions with sufficient detail to enable replication by future researchers.

6.2 Attention Collapse Diagnosis via XAI

The TFT’s failure at 8.48% MAPE was not immediately interpretable from the training loss curve alone—the loss converged smoothly with no gradient explosion signals. It was the application of the TFT’s built-in attention weight visualisation that revealed the pathology: the model was concentrating virtually all temporal attention on a single distant timestep ($t - 30$) across all prediction windows. This degenerate pattern indicates the model failed to learn meaningful temporal dependencies and instead latched onto a spurious long-range signal in the non-stationary price series.

This diagnosis guided the exact intervention: the logarithmic return transformation eliminates the long-run drift generating the false signal, forcing healthy attention redistribution

across the encoder window. Post-transformation attention patterns showed the expected decayed structure, with weights concentrated on recent $t - 5$ to $t - 10$ days—consistent with the known short memory of equity log-return series.

6.3 Data Starvation in Real-Time Inference

The TFT architecture requires a strict minimum encoder sequence length of 30 contiguous trading days. When the live trading engine fetched the trailing 60 calendar days via `yfinance`, it received approximately 28 usable trading days after accounting for NSE weekends, public holidays, and sporadic missing data—below the minimum threshold, causing a hard runtime crash during TFT dataset construction.

The resolution required engineering a dynamic time-delta buffer. Rather than requesting a fixed calendar window, the fetch function computes:

$$W_{\text{calendar}} = \lceil d_{\text{required}} \times \mu \rceil \quad (6.1)$$

where $d_{\text{required}} = 32$ (30-day minimum plus 2-day safety margin) and μ is a multiplier initialised at 2.5. After fetching, the pipeline counts valid trading rows; if the count remains below threshold, μ increments by 0.5 and the fetch retries. This iterative strategy robustly handles all NSE holiday patterns without hardcoding calendar-specific logic. To operationalize the dynamic time-delta buffer described in Equation 6.1, the following retry-loop logic was implemented within the live inference pipeline:

6.4 Cloud Infrastructure Volatility

Google Colab’s ephemeral runtime environment presents a fundamental challenge for persistent model deployment: any runtime disconnection results in complete loss of in-memory state, including loaded model weights. A naive implementation loading checkpoints from local Colab paths would require manual reloading on every session restart, breaking the autonomy of the automated trading engine.

The mitigation maps PyTorch Lightning’s `ModelCheckpoint` callback to write directly to a mounted Google Drive path at training time. The trading engine’s checkpoint loading function is hardcoded to the Drive path. On every fresh Colab session, the Drive mount

```
def fetch_trading_data(ticker, required_days=30, max_retries=5):
    multiplier = 2.5
    for attempt in range(max_retries):
        calendar_days = int(required_days * multiplier)
        start_date = today - timedelta(days=calendar_days)

        df = yf.download(ticker, start=start_date, end=today)
        df = df.dropna() # Remove weekends/holidays

        if len(df) >= required_days + 14: # +14 for RSI warmup
            return df

        multiplier += 0.5 # Expand search window if starved

    raise ValueError("Catastrophic data starvation: Failed to fetch.")
```

FIGURE 6.1: Dynamic Time-Delta Buffer Implementation for robust API ingestion.

is the first executed operation, and the model loads transparently from persistent storage. This architecture treats Google Drive as a lightweight model registry, enabling the trading engine to resume after any runtime interruption without human intervention.

6.5 Multi-Entity Data Alignment

Training a single model on five equities with differing trading volumes, listing histories, and holiday-specific missing data introduced a data alignment challenge. Misaligned timestamps corrupt the TFT's strict time-index requirements, causing silent errors where sequences from different entities are incorrectly paired during batching.

The mitigation enforces a unified trading calendar as the master index, computed as the intersection of valid trading days across all five equities and both macroeconomic series. Non-trading observations are forward-filled with a maximum gap constraint of one day (preventing spurious fill across extended holidays). Per-entity z-score normalisation is applied independently to prevent dominant high-price equities (Reliance) from overwhelming the gradient signal for lower-priced equities (ITC).

Chapter 7

Conclusion and Future Work

7.1 Summary of Contributions

This mid-semester report documents the successful completion of a multi-tier research sprint that has substantially extended the scope and depth of the project. The key contributions are:

1. **Probabilistic Forecasting Architecture:** A Temporal Fusion Transformer trained on stationary log-returns achieving 0.94% MAPE with calibrated 80% confidence intervals—a 47% improvement over the LSTM baseline.
2. **Global Multimodal Extension:** A cross-sectoral five-entity architecture with macroeconomic covariate injection that matches single-asset accuracy while generalising across fundamentally different return dynamics.
3. **XAI Discovery:** Formal quantitative proof that India VIX (>60% encoder importance) dominates individual stock Volume (~12%) as a predictor of short-horizon volatility—a contribution to the market microstructure literature.
4. **Counterfactual Validation:** Empirical demonstration that a 50% VIX spike expands the 80% confidence interval by over 300% while leaving the median prediction unchanged—proving the model encodes systemic risk into distributional uncertainty rather than directional bias.
5. **Live Trading Engine:** A deployed, automated paper trading system executing real-time probabilistic inference and risk-adjusted signal generation with a formally defined decision rule.

7.2 Limitations

Several constraints bound the current scope of findings. The paper trading engine has been operational for a limited period; no statistically significant conclusions about trading performance can yet be drawn. The counterfactual experiment is conducted on a single test window—a broader sweep across multiple market regimes would strengthen generalisability. The five-stock entity set does not represent the full diversity of Nifty 50 dynamics; mid-cap equities with different liquidity profiles may exhibit different XAI importance patterns. Additionally, the calibration of the confidence intervals has not been formally verified through conformal prediction coverage guarantees.

7.3 Future Work

- Expand the entity set to all 50 Nifty 50 constituents and compare XAI importance patterns across market-cap tiers and sector classifications.
- Integrate order book depth and FII/DII flow data as additional macroeconomic covariates to capture institutional trading pressure.
- Implement conformal prediction calibration to formally guarantee coverage probabilities of the confidence intervals.
- Conduct rolling backtests across multiple historical market regimes to evaluate risk-adjusted performance metrics (Sharpe ratio, maximum drawdown, Calmar ratio).
- Explore counterfactual analysis for other systemic risk events (RBI monetary policy announcements, geopolitical crude oil shocks) to build a broader causal map of the model’s risk-encoding behaviour.
- Investigate Normalising Flows or Diffusion Models as alternative probabilistic decoders to replace quantile regression, enabling richer and better-calibrated distributional outputs.

Appendix A

Hyperparameter Ablation Study

During the development of the Temporal Fusion Transformer, multiple architectural configurations were evaluated to optimize the bias-variance tradeoff. Table A.1 details the progression of hyperparameter tuning that led to the final Global Multimodal architecture.

Architecture	Hidden Size	Dropout	Target	Note	Test MAPE
Single TFT	32	0.10	Absolute	Severe Attention Collapse	8.48%
Single TFT	64	0.10	Log Return	High variance on test set	1.42%
Single TFT	32	0.10	Log Return	Optimal single-asset state	0.94%
Global TFT	64	0.10	Log Return	Underfitting multi-entity	1.88%
Global TFT	128	0.15	Log Return	Final Optimized Model	0.94%

TABLE A.1: Progression of hyperparameter tuning across model iterations.

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